Experiment No. 10

Version Control using Git & GitHub

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Roll No:	72		

Aim

To understand and implement Version Control using Git and GitHub by creating repositories, committing changes, and pushing code to a remote repository.

Theory

Git is a distributed version control system that tracks changes in source code during software development. It allows multiple developers to work together on non-linear development.

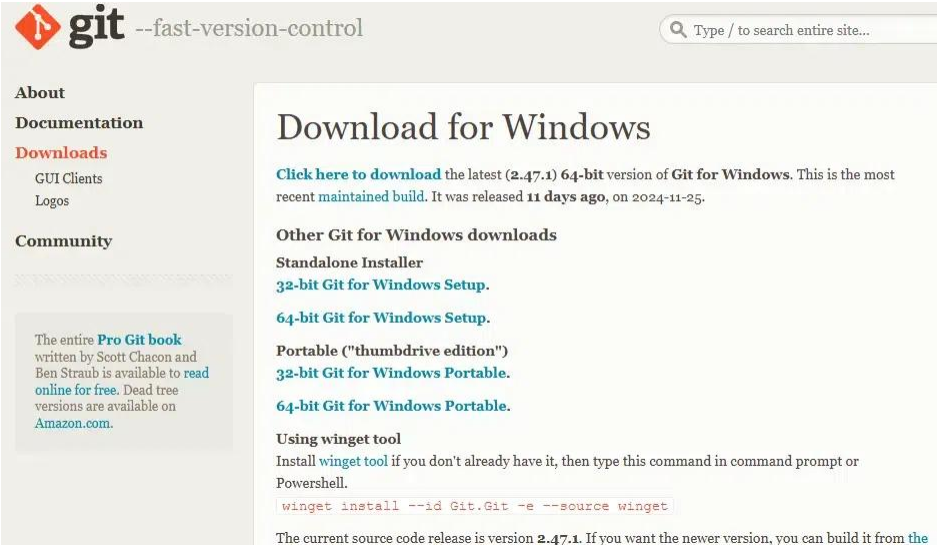
GitHub is a cloud-based hosting service that lets you manage Git repositories. It provides a web interface for collaboration, code review, and project management.

Key concepts include: Working Directory, Staging Area (Index), Local Repository, and Remote Repository. Changes flow from working directory → staging area → local repository → remote repository.

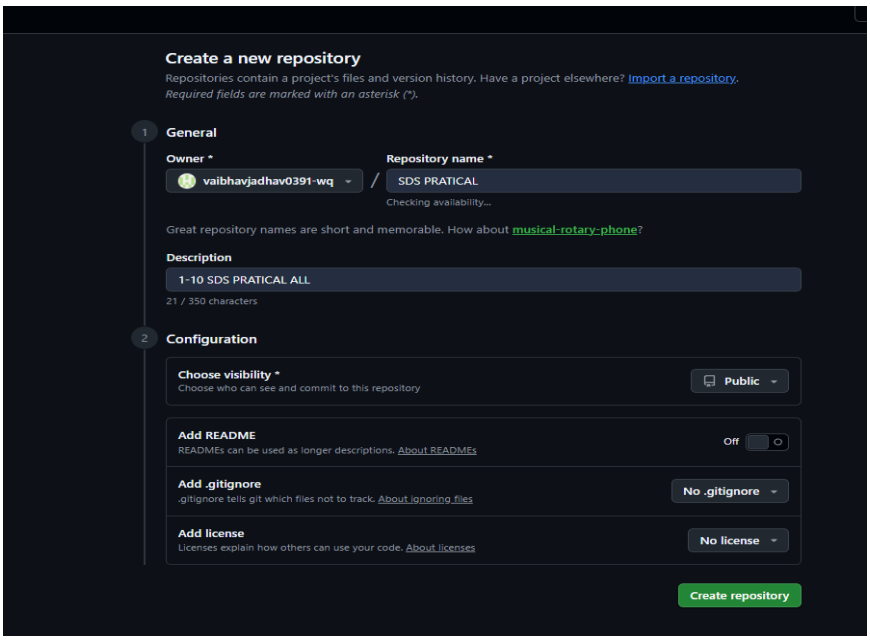
Step by Step Procedure

Sr.	Step Description
1.	Installation of Git
2.	Configure username & email
3.	Create local repository
4.	Create files
5.	Add files to staging
6.	Commit changes
7.	Create GitHub repository
8.	Link local repo to GitHub
9.	Push files

Screenshots



1. installation verification (git version)



2. Repository creation

MINGW64:/e/sds/vaibhav

```
Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds (master)
$ ls /e
 123/                                                      Music/                                          'doc submit to mit'/
 6TH/                                                      Notifications/                                   honar/
 A/                                                        Objective.docx                                  'micro project final'/
 AJP-classTest2.docx                                       Pictures/                                        miro/
'AVANTIAKA JADHAV'/                                        Podcasts/                                        mit/
 Alarms/                                                   Presentation2.pptx                               ppt/
 Android/                                                  Recordings/                                     'ppt no 2&3.pptx'
'Animated ppt 2.pptx'                                      Ringtones/                                       project/
 Audiobooks/                                               STE/                                             sds/
'Border-2.2026.1080p.V2-HDTC.Hindi-Line.x264-HDHub4u.Ms (2).mkv'  'System Volume Information'/             'shivani py.docx'
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'Final Year Project Dairy for 6th Semester.rtf'            WirelessLan_ROG_Realtek_B_V6001.15.160.100Sub1_44703.exe*  '~$Animated ppt.pptx'
'Final Year Project report first pages.pdf'                Word.lnk*                                       '~$Mp.pptx'
'Java program'/                                            aikr/                                           '~$Presentation2.pptx'
 LOST.DIR/                                                 aikr.zip                                        '~$Wood Type.pptx'
'MAD Project'/                                             cpe.pdf                                         '~WRL0003.tmp'
'Microproject Report Format (4).docx'                      cpe.rtf                                         '~WRL1068.tmp'
 Movies/                                                   cpp/                                            ''$'\360\237\230\201'' VAIBHAV DOC'/
 Mp.pptx                                                  'cpr mit.docx'                                   ''$'\360\237\247\221\342\200\215\360\237\222\273''*'/

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds (master)
$ cd/e
bash: cd/e: No such file or directory

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds (master)
$ cd /e
ls
 123/                                                      Music/                                          'doc submit to mit'/
 6TH/                                                      Notifications/                                   honar/
 A/                                                        Objective.docx                                  'micro project final'/
 AJP-classTest2.docx                                       Pictures/                                        miro/
'AVANTIAKA JADHAV'/                                        Podcasts/                                        mit/
 Alarms/                                                   Presentation2.pptx                               ppt/
 Android/                                                  Recordings/                                     'ppt no 2&3.pptx'
'Animated ppt 2.pptx'                                      Ringtones/                                       project/
 Audiobooks/                                               STE/                                             sds/
'Border-2.2026.1080p.V2-HDTC.Hindi-Line.x264-HDHub4u.Ms (2).mkv'  'System Volume Information'/             'shivani py.docx'
'CSS MICROPROJECT'/                                        'This PC - Shortcut.lnk'*                       'ste 2'/
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'Final Year Project report first pages.pdf'                Word.lnk*                                       '~$Mp.pptx'
'Java program'/                                            aikr/                                           '~$Presentation2.pptx'
 LOST.DIR/                                                 aikr.zip                                        '~$Wood Type.pptx'
'MAD Project'/                                             cpe.pdf                                         '~WRL0003.tmp'
'Microproject Report Format (4).docx'                      cpe.rtf                                         '~WRL1068.tmp'
 Movies/                                                   cpp/                                            ''$'\360\237\230\201'' VAIBHAV DOC'/
 Mp.pptx                                                  'cpr mit.docx'                                   ''$'\360\237\247\221\342\200\215\360\237\222\273''*'/

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e
$ cd sds
```

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MINGW64:/e/sds/vaibhav

```
Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e
$ cd sds

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds (master)
$ cd vaibhav

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git init
Initialized empty Git repository in E:/sds/vaibhav/.git/

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav
$ git add .
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

        git config --global --add safe.directory E:/sds/vaibhav

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav
$ git commit -m "Added PDFs"
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

        git config --global --add safe.directory E:/sds/vaibhav

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
fatal: detected dubious ownership in repository at 'E:/sds/vaibhav'
'E:/sds/vaibhav' is on a file system that does not record ownership
To add an exception for this directory, call:

        git config --global --add safe.directory E:/sds/vaibhav

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav
$ git config --global --add safe.directory E:/sds/vaibhav

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git add .

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git commit -m "Added PDFs"
[master (root-commit) e563d1b] Added PDFs
 9 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 84-72-01-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-02-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-03-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-04-VAIBHAV JADHAV (2).pdf
 create mode 100644 84-72-05-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-06-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-07-VAIBHAV JADHAV.pdf
 create mode 100644 84-72-08-VAIBHAV JADHAV.pdf
 create mode 100644 report.pdf

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
```

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```
 create mode 100644 B4-72-08-VAIBHAV JADHAV.pdf
 create mode 100644 report.pdf

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git remote add origin https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ ^[[200~git branch -M main
bash: $'\E[200~git': command not found

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (master)
$ git branch -M main

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$ git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (11/11), 805.41 KiB | 14.13 MiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vaibhavjadhav0391-wq/SDS-PRATICAL-.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$
```

3. Successful push

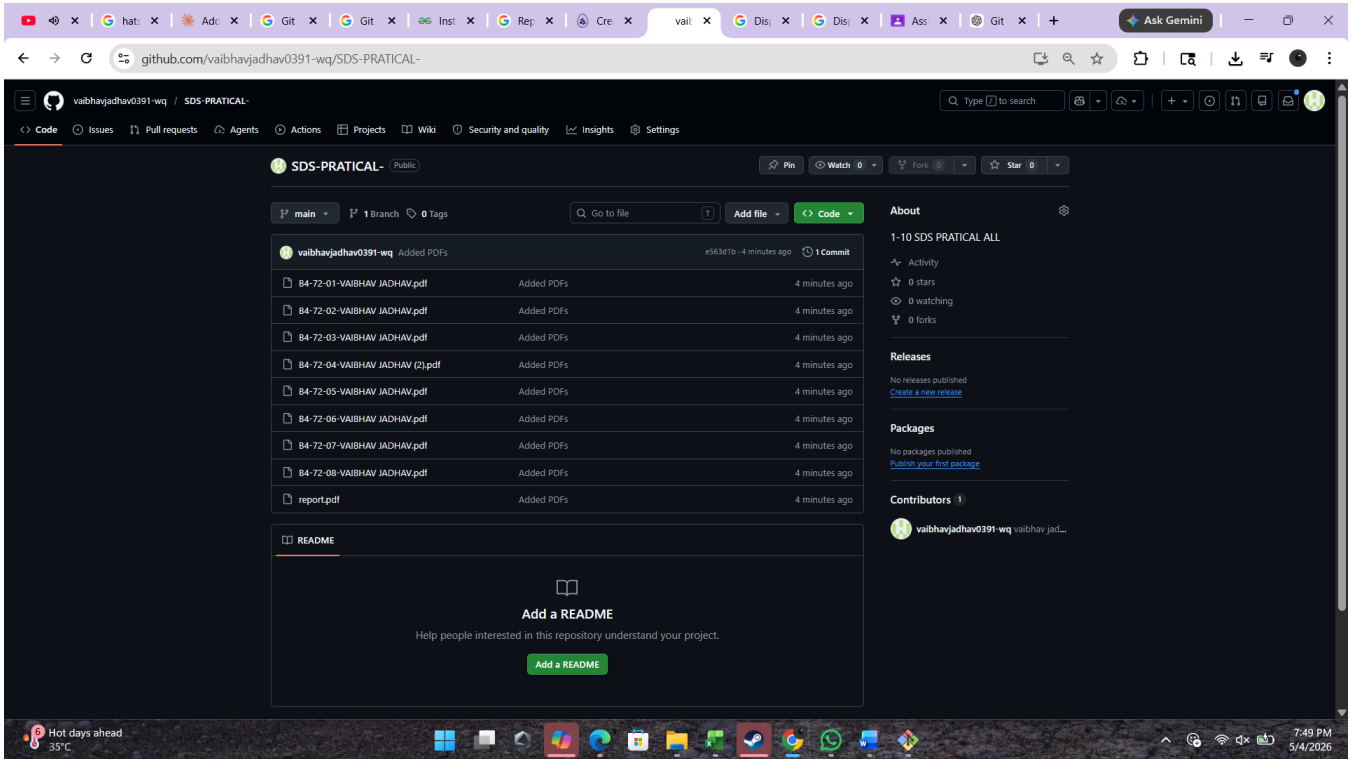
```
branch 'main' set up to track 'origin/main'.

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$ git log
commit e563d1b8754b93348623ba45bacfad1411ea4b1d (HEAD -> main, origin/main)
Author: Vaibhav Jadhav <vaibhavjadhav0391@gmail.com>
Date:   Mon May 4 19:45:20 2026 +0530

    Added PDFs

Vaibhav Jadhav@LAPTOP-MEJOP5TN MINGW64 /e/sds/vaibhav (main)
$
```

4. Display commit logs (git log)



5. Show files on GitHub repo

Questions

1. What is the difference between Git and GitHub?

Git and GitHub are related but different tools used in software development.

- **Git is a distributed version control system. It is used to track changes in files and manage different versions of a project locally on a computer. It allows developers to work offline and maintain history of changes.**
- **GitHub is a cloud-based platform that hosts Git repositories. It provides features like collaboration, sharing code, issue tracking, and version management over the internet.**

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2. Explain the difference between git add and git commit

- **git add:**
  This command is used to move changes from the working directory to the staging area. It prepares the files for saving but does not actually save them.
- **git commit:**
  This command saves the staged changes permanently in the repository. Each commit creates a snapshot of the project with a message describing the changes.

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3. What is the staging area in Git?

The staging area is an intermediate area in Git where changes are stored before they are committed.

- **It allows the user to select specific changes to include in the next commit**
- **It acts as a buffer between working directory and repository**
- **It helps in organizing commits properly**

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4. What happens if you modify a file after committing it?

If a file is modified after committing:

- **The new changes are not automatically saved in the repository**
- **Git marks the file as "modified"**
- **You must again use git add and git commit to save the new changes**

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5. Difference between git push and git pull

- **git push:**
  It is used to upload local commits from your system to a remote repository like GitHub. It updates the online repository.

- **git pull:**
  It is used to fetch and download changes from the remote repository to your local system. It keeps your project updated.

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6. What is Version Control? Differentiate between Centralized and Distributed VCS

Version Control is a system that records changes made to files over time so that previous versions can be restored when needed.

Centralized Version Control System (CVCS):

- **Has a single central server**
- **All users connect to this server**
- **Requires internet connection**
- **If server fails, data may be lost**
- **Example: SVN**

Distributed Version Control System (DVCS):

- **Every user has a complete copy of the repository**
- **Works offline**
- **Faster and more secure**
- **No single point of failure**
- **Example: Git**

Conclusion

Git and GitHub are powerful tools used for managing and sharing projects. Git helps in tracking changes and maintaining version history, while GitHub allows collaboration and remote storage. Version control systems improve productivity, ensure data safety, and make teamwork easier in sof